

PORT VILA EFATE ISLAND, Vanuatu

WMO: 915570

Lat: 17.699S

Lon: 168.320E

Elev: 21

StdP: 101.07

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	15.8	13.1	9.4	18.6	14.1	10.0	18.7	9.2	24.5	8.4	24.6	0.7	60	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.0	31.7	26.4	31.1	26.1	30.6	25.9	27.4	30.2	27.0	29.9	26.7	29.5	3.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.3	29.1	26.2	21.6	28.8	25.9	21.2	28.5	87.1	30.3	85.2	29.9	83.7	29.8	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.6	7.7	7.0	DB	12.7	32.7	1.3	0.7	11.8	33.3	11.0	33.7	10.2	34.1	9.3	34.6
			WB	12.6	28.4	1.4	0.8	11.6	29.0	10.8	29.5	10.0	29.9	8.9	30.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	26.6	26.9	26.7	25.8	24.2	23.1	22.1	22.1	22.9	23.9	24.8	25.9
	DBStd	2.23	1.09	1.05	0.90	1.14	1.45	1.54	1.70	1.88	1.80	1.59	1.41	1.28
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5319	515	474	518	473	439	392	376	376	387	430	446	494
	CDD18.3	2277	257	241	259	223	181	142	117	118	137	172	196	236
	CDH23.3	17372	2489	2361	2423	1782	1149	647	445	502	753	1156	1513	2151
	CDH26.7	4168	781	759	708	407	154	49	32	36	89	195	329	629
Wind	WSAvg	2.6	2.4	2.3	2.2	2.2	2.4	2.4	2.4	2.8	2.8	3.2	3.0	2.7
Precipitation	PrecAvg	2249	329	291	389	220	161	152	92	115	106	123	156	179
	PrecMax	3345	632	798	831	494	460	429	224	335	336	566	511	566
	PrecMin	1091	58	73	115	60	49	20	7	13	12	16	16	25
	PrecStd	507	159	152	175	129	103	97	64	84	78	114	118	130
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	33.0	32.1	31.4	30.0	29.2	28.9	29.0	30.0	30.6	31.0	32.0
		MCWB	26.5	27.3	26.5	26.4	25.6	25.1	24.5	24.6	24.6	25.0	25.3	26.0
	2%	DB	31.3	31.7	31.2	30.4	29.0	28.0	27.6	27.7	28.7	29.3	30.1	31.0
		MCWB	26.2	26.7	26.5	26.1	25.0	24.2	23.7	23.6	23.9	23.9	24.9	25.6
	5%	DB	30.7	31.0	30.6	29.7	28.2	27.1	26.6	26.8	27.7	28.5	29.2	30.2
		MCWB	25.9	26.3	26.3	25.7	24.3	23.6	22.9	22.8	23.2	23.5	24.4	25.4
	10%	DB	30.1	30.2	29.9	29.0	27.5	26.2	25.7	25.9	26.8	27.8	28.6	29.7
		MCWB	25.6	26.0	26.0	25.4	23.9	23.0	22.2	22.1	22.5	23.1	24.0	25.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	28.2	27.7	27.2	26.7	25.8	25.6	25.2	25.6	26.0	26.6	27.2
		MCDB	30.8	31.3	30.1	29.5	28.2	28.0	27.8	27.9	28.3	29.3	29.7	30.3
	2%	WB	27.1	27.4	27.2	26.7	25.9	25.1	24.5	24.3	24.7	25.2	25.9	26.6
		MCDB	30.1	30.5	30.0	29.2	28.0	27.0	26.6	26.6	27.5	28.0	28.8	29.7
	5%	WB	26.6	27.0	26.7	26.3	25.3	24.5	23.7	23.7	24.1	24.6	25.4	26.2
		MCDB	29.5	30.0	29.6	28.7	27.3	26.3	25.7	25.7	26.7	27.2	28.1	29.2
	10%	WB	26.2	26.5	26.4	25.9	24.8	23.9	23.1	23.1	23.5	24.0	24.9	25.8
		MCDB	29.0	29.4	29.1	28.3	26.7	25.7	24.9	25.1	25.9	26.5	27.5	28.6
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.0	6.6	6.6	7.0	6.5	7.0	7.3	7.8	7.8	7.5	7.6
		MCDBR	8.1	8.0	7.7	7.6	7.7	7.4	7.9	7.9	8.6	8.9	8.7	8.5
		MCWBR	3.8	3.8	3.8	4.0	4.2	4.2	4.5	4.3	4.5	4.3	4.0	4.0
	5% WB	MCDBR	7.0	7.2	7.0	6.7	6.4	6.2	6.6	6.3	7.1	7.0	7.2	7.2
		MCWBR	3.7	3.8	3.8	3.8	4.0	4.0	4.2	3.8	4.2	3.8	4.0	3.8
Clear-Sky Solar Irradiance	taub	0.402	0.394	0.389	0.386	0.389	0.393	0.383	0.380	0.388	0.388	0.395	0.396	0.390
	taud	2.557	2.591	2.616	2.618	2.573	2.552	2.557	2.563	2.524	2.530	2.546	2.577	
	Ebn at Noon	943	941	924	890	850	827	847	883	909	929	942	954	
	Edh at Noon	109	104	98	92	91	89	91	96	106	110	110	107	
All-Sky Solar Radiation	RadAvg	6.41	6.17	5.57	4.75	4.04	3.60	3.81	4.46	5.41	5.93	6.37	6.62	
	RadStd	0.57	0.43	0.45	0.40	0.38	0.27	0.27	0.32	0.48	0.47	0.52	0.41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	+0.33	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	+0.33	N/A	N/A	N/A

Nomenclature: See separate page